

LOKESH VARMA KONDURU

(+1) 571-244-4062 · lokesh.v.konduru@gmail.com · [linkedin.com/in/lokeshvarmakonduru](https://www.linkedin.com/in/lokeshvarmakonduru) · github.com/LokeshVarma-Konduru

SUMMARY

Full-Stack Software Engineer with **3+ years** of experience specializing in **Distributed Systems, Real-Time Architectures, and Cloud Infrastructure**. Proven track record delivering production-grade applications and backend microservices across Agri-tech, fraud detection, and e-commerce domains. Expert in **React/TypeScript, Node.js/Express, Python/FastAPI, and AWS/Kubernetes**, integrating **AI-powered features** into production systems. M.S. Computer Science, **Virginia Tech (GPA: 4.0/4.0)**.

EDUCATION

Virginia Tech — M.S. in Computer Science GPA: 4.0 / 4.0 | Aug 2024 – May 2026
Mahindra University — B.Tech. in Computer Science GPA: 8.28 / 10 | Jul 2019 – May 2023

PROFESSIONAL EXPERIENCE

Software Engineer **Virginia Tech** *Sep 2024 – May 2026*

- Cut **bug turnaround time by 40%** by building a full-stack platform with **React/TypeScript, Next.js SSR**, and PostgreSQL, backed by Jest unit and integration test coverage.
- Reduced deployment time by **65%** by containerizing services with **Docker and Kubernetes** on AWS EC2 and automating CI/CD pipelines via GitHub Actions with Terraform IaC.
- Improved issue resolution **by 20%** by shipping 5+ geospatial analytics tools using **Google Earth Engine APIs** on a React/Tailwind platform with Firebase telemetry and **Prometheus/Grafana** observability. [Website](#) | [VT Publications](#)
- Boosted geospatial data processing efficiency **by 40%** by building an async **Python/FastAPI** backend with **Redis caching** and PostgreSQL, improving overall pipeline throughput.
- Engineered an **AI-powered advisory assistant** by integrating **LangChain, LangGraph, and FAISS** into Node.js/Express REST APIs, implementing RAG pipelines and MCP tool calling to deliver real-time contextual recommendations.

Software Engineer (Contract) **Marine Corps Community Services (MCCS)** *Sep 2024 – Dec 2025*

- Scaled fraud detection to handle **10K+ transactions** by architecting microservices with **Node.js/Express, MongoDB, and Redis** for real-time fraud score and transaction lookups.
- Reduced **fraud signal processing latency by 40%** by orchestrating **Kafka event streaming on Kubernetes** with **CloudWatch and OpenTelemetry** distributed tracing, integrating SageMaker for ML inference and S3 for model artifact storage. [GitHub](#)
- Reduced manual fraud review time by **30%** by building a **GenAI fraud investigation assistant** with **LangChain, LangGraph, and FAISS**, implementing RAG pipelines and MCP tool calling for intelligent fraud signal analysis.

Software Engineer **DuoDecimal Tech LLP** *Jul 2023 – Jul 2024*

- Delivered **4 production platforms** — events, bookings, hostel, and food management — using **React/TypeScript, Node.js/Express, and MongoDB** with JWT/OAuth2 authentication.
- Improved booking efficiency **by 30%** and cut response latency **by 50%** by building a real-time ordering platform with **Socket.io and Kafka**, integrating Razorpay payment processing and Redis-backed session caching.
- Reduced manual paperwork **by 60%** for **25,000+ users** by developing an RFID-integrated hostel management system with **Node.js/Express and MongoDB** indexed queries and event-driven access control.
- Enabled zero-downtime releases for **25,000+ users** by deploying microservices on **AWS EC2 behind ALB**, routing through API Gateway with GitHub Actions CI/CD and automated Jest testing.

Software Engineer Intern **MindGraph Technologies** *Jan 2023 – Jun 2023*

- Improved data visibility **by 40%** for Telecom and CPG clients by building dashboards with **Python/Flask and Plotly Dash**, integrating Azure AD SSO and deploying on **Microsoft Azure** with DevOps CI/CD.
- Ensured data quality and reliability across **10M+ airline records** by designing distributed pipelines with **PySpark, Kafka, and Apache Airflow**, with automated PyTest unit and integration testing.

PROJECTS

Nova Clinic | *Python/FastAPI, MySQL, HTML5/CSS, JWT* *Feb 2026 – May 2026*

- Achieved **high availability with zero downtime** during production deployment by building JWT-authenticated REST APIs over HTTPS with Python/FastAPI and MySQL, with a responsive HTML5/CSS frontend for appointment management. [GitHub](#)
- Designed a normalized **MySQL schema** with role-based access control, enforcing data privacy through JWT claim-based permissions.

E-Commerce Bookstore Platform | *React, Java/Spring Boot, MySQL, Redis* *Feb 2025 – May 2025*

- Delivered catalog, cart, and checkout workflows for a full-stack bookstore using **Java/Spring Boot, Spring Security, and MySQL** with JWT/OAuth2 authentication, achieving sub-200ms API response times.
- Improved recommendation accuracy for a **200+ book catalog** by integrating a **LangChain-powered recommendation assistant** with **FAISS** semantic search and **Redis** session caching to support personalized user flows.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL, C, C++, HTML5, CSS
Frontend: React.js, Next.js SSR, Redux Toolkit, React Query, Tailwind CSS, HTML5/CSS3, Socket.io, Plotly Dash
Backend: Node.js/Express, Java/Spring Boot, Spring Security, FastAPI, Flask, REST APIs, Microservices, Distributed Systems, JWT/OAuth2, pandas, NumPy, PySpark
Databases & Caching: PostgreSQL, MongoDB, MySQL, Redis, Firebase
AI Integration: LangChain, LangGraph, OpenAI API, RAG Pipelines, FAISS, MCP Tool Calling
Cloud & DevOps: AWS (EC2, S3, SageMaker, API Gateway, ALB, CloudWatch), Kubernetes, Docker, Kafka, Terraform, GitHub Actions CI/CD, Apache Airflow, Microsoft Azure, Azure DevOps, OpenTelemetry, Prometheus/Grafana, Linux
Testing & Tools: Jest, PyTest, TDD, Git, Agile/Scrum, Jira, System Design, Google Earth Engine